Synthetic Data Generation

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Part 1

Introduction

Background

In today’s digital world, **data is everywhere**.

Every step in modern life is powered by data — from mobile payments and navigation apps, to online shopping and social media feeds.

Our habits, preferences, and digital footprints are constantly collected and analysed to personalise what we see, buy, and use.

Everything runs on data.
Decisions, predictions, and innovations are all built upon it.

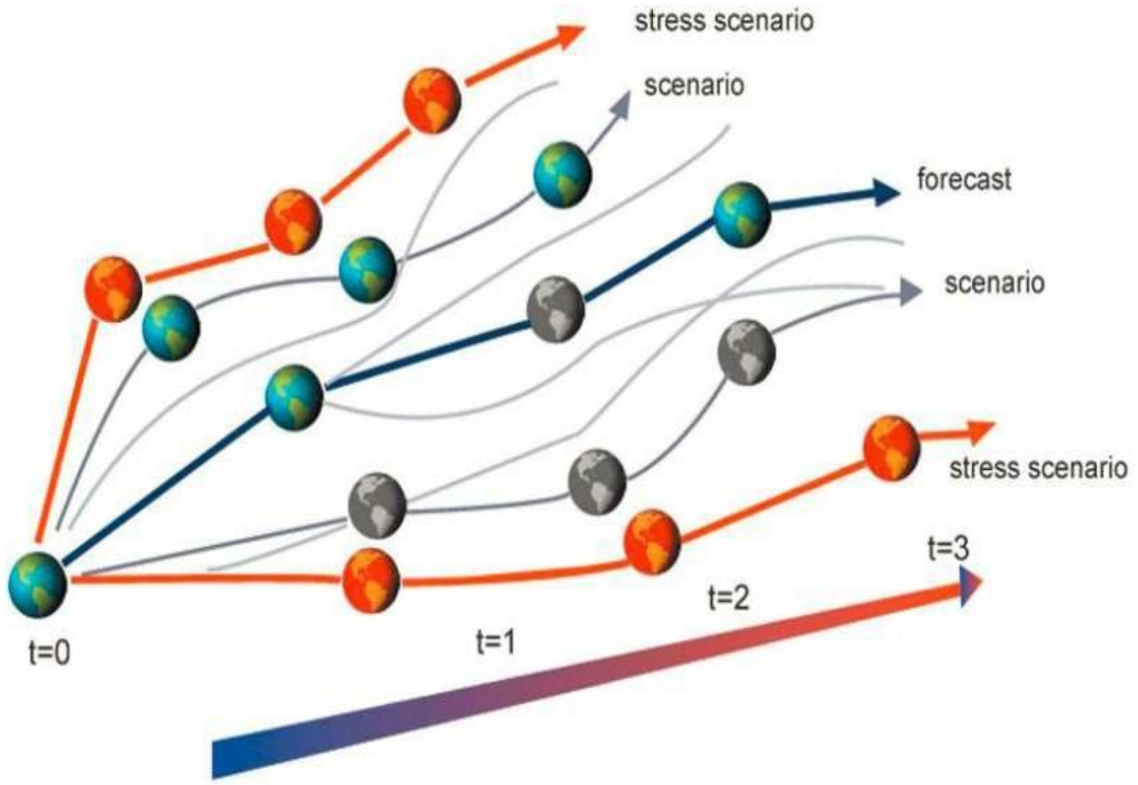
But **real-world data has its limits**

Challenges with real datasets

Coverage

- It may not be feasible to get samples for all categories
- Lighting conditions
- Modifications (Glasses/No glasses, Moustache/ No Moustache etc.)
- Positions

Challenges with real datasets



All scenarios haven't played out

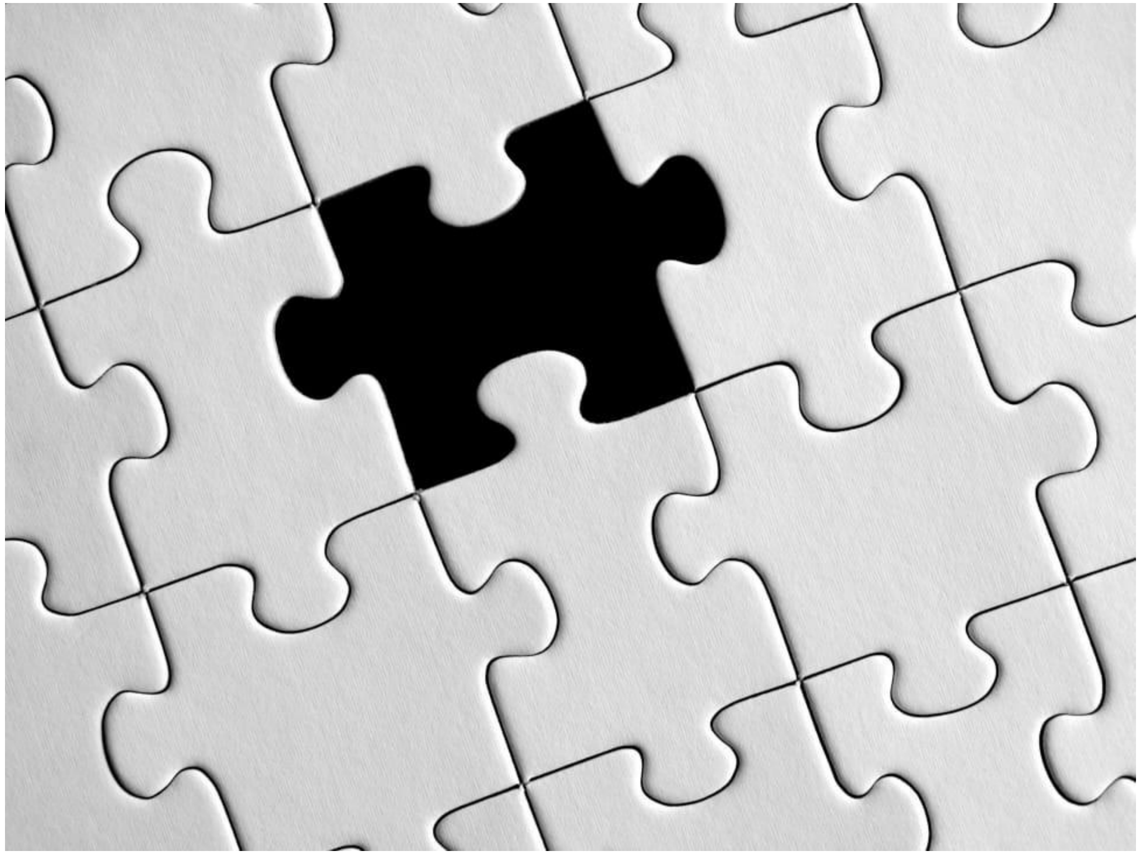
- Stress scenarios
- What-if scenarios

Figure ref: http://www.actuaries.org/CTTEES_SOLV/Documents/StressTestingPaper.pdf

Challenges with real datasets

Missing values

- Missing at random
- Missing sequences
- Need data to fill frames



Challenges with real datasets

Access

- Hard to find
- Rare class problems
- Privacy concerns making it difficult to share

Challenges with real datasets

Imbalanced

- Need more samples of rare class
- Need proxies for data points that were not observed or recorded

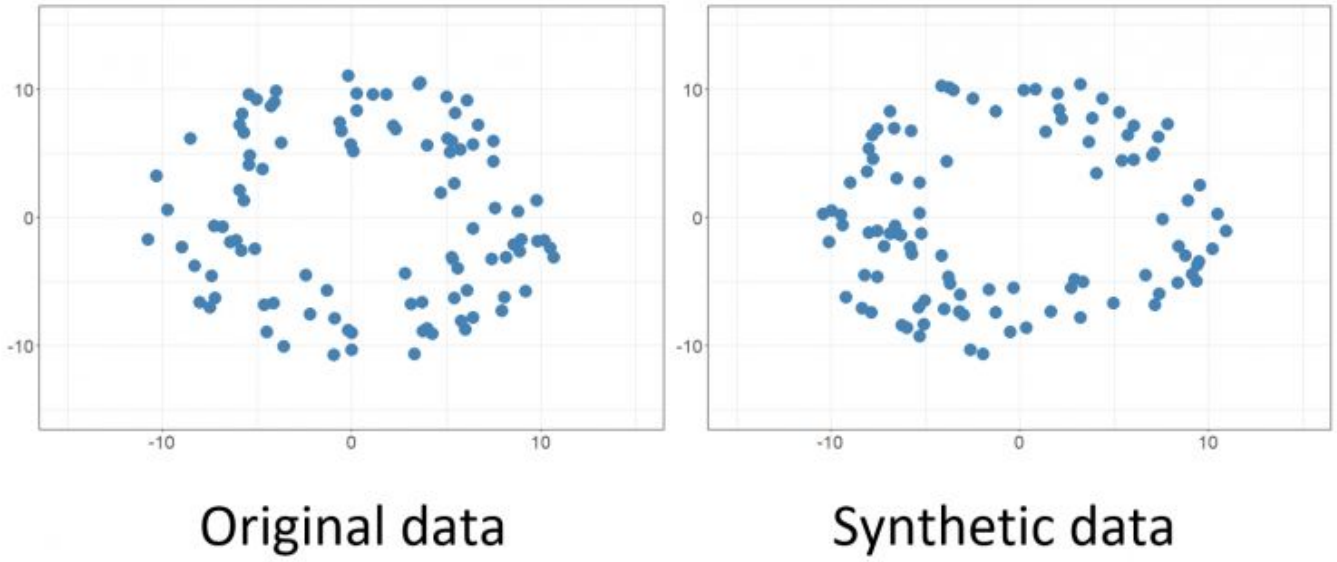
Challenges with real datasets

Labels

- Human labeling is hard
- Synthetic label generators

What is synthetic data

Synthetic data is **artificially generated data** that is not collected from real world events. It **replicates the statistical components** of real data **without containing any identifiable information, ensuring individuals' privacy.**



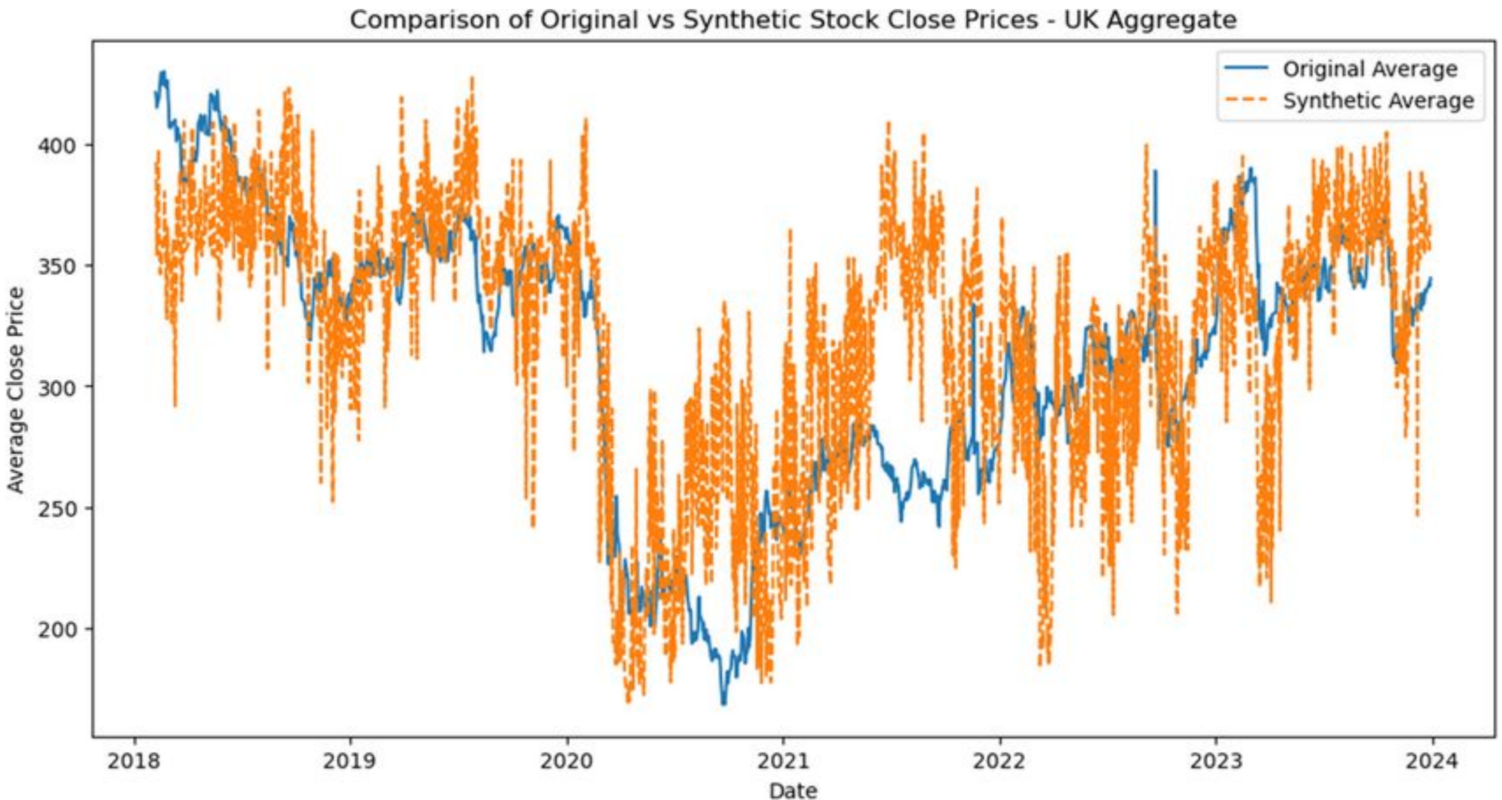
The synthetic data retains the structure of the original data but is not the same

Types of Synthetic Data

- Structured Data
- Unstructured Data
- Sequential Data

Sequential data: involves time or order, making it essential for models that need to understand temporal relationships. Examples include:

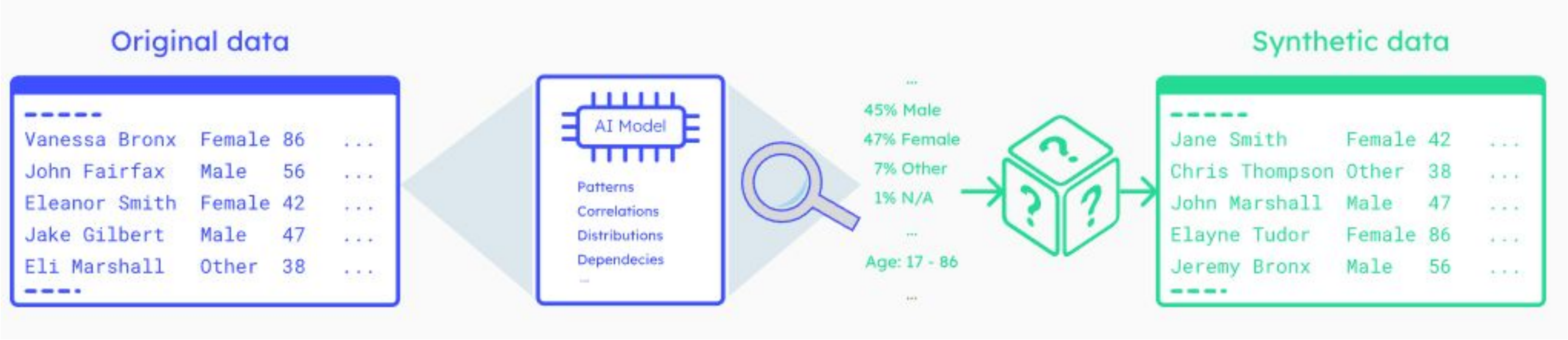
Time-series data: Synthetic stock prices, weather patterns, or energy consumption data.



Structured data: databases and spreadsheets

Customer records: Synthetic customer profiles with attributes like name, age, address, and purchase history.
Transaction logs: Artificially generated financial transactions or sales data.
Sensor readings: Simulated data from IoT devices or industrial sensors.
Healthcare records: Synthetic patient data, including diagnoses, treatments, and outcomes.

Generation methods for structured synthetic data often involve statistical modeling, rule-based systems, or machine learning techniques like variational autoencoders (VAEs) or generative adversarial networks (GANs).



Unstructured data

Images: Generative AI produced images from tools such as Stable Diffusion, Midjourney, DALL-E, etc.
Audio files: Synthetic speech, music, or sound effects usually produced using Generative AI algorithms.
Text documents: Artificially generated text using GenAI algorithms such as GPT-4, Claude, LLaMA, Mixtral, etc.
Video: Synthetic video footage produced using GenAI algorithms like OpenAI SoRA.

Bounding box labels from human labelers *Ground truth labels from synthetic data*

Theme and use cases

I want to build a robot that picks aluminum cans out of the trash. These cans can crumple in thousands of different ways. If I have 10 pictures of crumpled Coca-Cola cans, that may not be enough to train a machine learning model to identify cans at a high enough rate. I need more data. One option is to go crumple 5,000 more cans, in all different ways, in all different weather, laying next to all kinds of different things in a landfill. That’s expensive. The easier option is to take a physics engine and create 5,000 fake crumpled can images to add to my data set, that are based on the real crumpled can.

Theme and use cases

Theme 1: Data augmentation and bias mitigation

Use Case Title	Focus
1. Transaction sequences used in fraud detection machine learning models	The availability of fraudulent financial transaction data to train a fraud detection machine learning model
2. Reject inference in credit scoring	Generating synthetic data to mitigate selection bias in credit scoring training data

Theme and use cases

Theme 2: Systems testing and model validation

Use Case Title	Focus
1. Synthetic Open Banking data for model testing	Generating synthetic transaction data labels to enhance training sets and transactional data to resemble consumer patterns and behaviours
2. Cross-sector synthetic data for Authorised Push Payment Fraud	Creation of synthetic data relating to individual, banks and fraud typologies to complement real data and be used in the FCA's APP Fraud TechSprint

Theme and use cases

Theme 3: Internal and external data sharing

Use Case Title	Focus
1. Common data sets for community research into societal challenges	The US-UK Challenge using Privacy Enhancing Technologies to enable data sharing to improve financial crime prevention and pandemic responses
2. Data sharing to increase efficiency and effectiveness of anti-money laundering controls	Using cross-organisational and international transaction data to develop machine learning models to recognise illicit payment patterns

Theme and use cases

For some areas though, synthetic data works less well. Given 100 sentences of English, I may not be able to accurately generate 10,000 synthetic English sentences to put into a NLP model. The sentences may be very basic and not helpful to the model. Knowing when you can use synthetic data and when you can't is important.

Why we use it?

- Overcoming data scarcity
- Enhancing data privacy
- Better machine learning models
- Cost efficiency

Disadvantages

1. Regulatory/legal input:

There are regulatory considerations associated with synthetic data creation and usage, which vary according to the methodology, type of data and region (such as UK GDPR).

2. Dependency on real world data & Biases in generated data

The quality and accuracy of real-world data can impact the synthetic data being generated. Biases and inaccuracies in real world data can be replicated in synthetic datasets. Data and model testing and assessments are needed once the data have been generated to help address any biases that are present.

3. Methodology:

It is still an open question regarding what generation methods and combinations of real-life and synthetic data are optimal for model accuracy, ethically desirable and aligned with regulatory frameworks.

4. Validating the data:

Acceptable levels of privacy, utility and fidelity of the data should be considered on a case-by-case basis and according to the deployment of the data.

Part 2

Tools for Synthetic Data Generation

Proprietary Tools

Company	Core Technology
Tonic.ai	All-in-one platform for data anonymization, subsetting, and synthesis integrated with databases (hadoop, oracle, mysql, MS sql server, mongo db, amazon aurora/redshift, and google big query) - Uses Condenser and Masquerade
Mostly.ai	**Tablular data** using generative deep neural networks (no image data)
CVEDIA	-Sensor modeling and algorithm training -**Handle image** using SynCity as a custom pocket laboratory to generate highly entropic scenes, conditions, and metadata. Enable real-time Hardware-In-the-Loop (HWIL), Human-In-the-Loop (HITL) or Software-In-the-Loop (SIL) simulations even with complex sensor configurations
Deep vision data	**image creation** synthetic training data
Synthesis.ai	The data generation platform for **computer vision**

SDV

SDV is a **Python library**. It provides a library of models that can learn the structure and relationships in real data and then generate new synthetic samples that look statistically similar.

It supports various data types, including **time series**, **relational data**, and **tabular data**.

SDV uses advanced probabilistic models like **Gaussian copulas** and **deep learning methods** to learn the structure of the original data and produce high-quality synthetic versions.

It's widely used in machine learning applications for generating synthetic training data **when real data is limited or sensitive**.

```
pip install sdv
```

Gaussian Copula

Gaussian Copula is a statistical model used to generate synthetic tabular data. It works by modeling each column’s individual distribution, and then using a multivariate Gaussian structure to capture the correlations between columns.

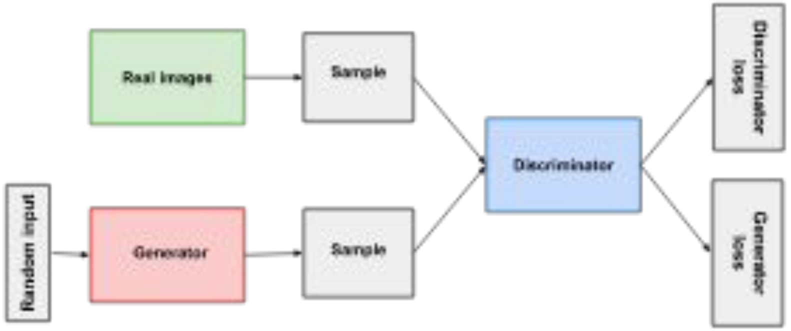
Best use cases:

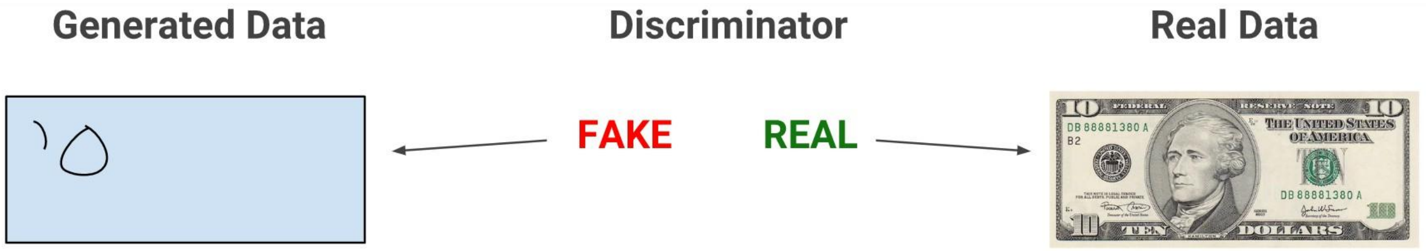
- When data is structured, numeric, and not too complex.
- Works very well for finance, demographics, or risk modeling, where relationships between columns are roughly linear or monotonic.
- Also great when you need a transparent, explainable model.

Limitations:

- It assumes Gaussian (linear) correlations, so it struggles with nonlinear or multimodal relationships.
- It doesn’t handle categorical or mixed-type variables as naturally as deep learning models.

GAN





As training progresses, the generator gets closer to producing output that can fool the discriminator:

Finally, if generator training goes well, the discriminator gets worse at telling the difference between real and fake. It starts to classify fake data as real, and its accuracy decreases.

https://developers.google.com/machine-learning/gan/gan_structure

CTGAN (Conditional Tabular GAN)

it’s a deep learning model based on Generative Adversarial Networks (GANs), specifically designed for tabular data.

How it works:

- It has a Generator that creates fake data and a Discriminator that tries to detect whether the data is real or synthetic.
- The two networks compete until the generator learns to produce data that the discriminator can’t tell apart from real data.
- The “Conditional” part means it can generate data conditioned on categorical values, like ‘room_type = Deluxe’.

Best use cases:

- When data includes **both numerical and categorical variables**.
- When there are nonlinear or complex dependencies between features.
- Ideal for business, healthcare, and financial datasets where relationships are intricate and not easily captured by traditional statistics.

Limitations:

- Harder to train — it requires tuning and more computational power.
- Less interpretable — deep networks are black boxes.
- May overfit small datasets or create unrealistic outliers if not trained carefully.

https://docs.sdv.dev/sdv/tutorials

https://colab.research.google.com/drive/1MCTkTj9-93Ei-cLDQoj9AXaqPhpue7a3?usp=sharing#scrollTo=qdseltjrFuMO

https://colab.research.google.com/drive/15iom9fO8j_gHg4-NIGkzWF5thMWStXwv?usp=sharing#scrollTo=UIHA32qMIIqw

Thanks